Giacomo Veneri, MEng, PhD

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LinkedIn: **https://www.linkedin.com/in/giacomoveneri**

Github **:** **https://github.com/venergiac**

Knowledge Panel**:** **https://share.google/IGMbOuBNOKy1eQrr6**

SUMMARY

Experienced leader with a strong background in digital services, **AI, and IoT applied to Industry (Energy, Transportation, Manufacturing, Healthcare)**. Proven track record in developing and implementing innovative solutions (8 patents, 50+articles, 3 books, 70+ projects), leading global teams, and **supporting sales and commercial functions** (approx. 100M$+ orders / year). Leding with a collaborative and visionary approach, empowering teams through clear communication, trust, and a focus on innovation and continuous improvement Skilled in project management, financial analysis, **data** strategy, and navigating complex multinational environments. **Product** Management and Budget Management **SAFE Agile, NVIDIA HPC, Java, C3.ai, Data Bricks and Six Sigma Certified.**

EXPERIENCE

AI and IOT Fellow (Strategic Technology Program) - Feb 2026 – Present

Baker Hughes, Florence, Italy

- Define and execute the enterprise AI & IoT strategy, ensuring clear alignment with business priorities, growth objectives, and digital transformation initiatives.
- Drive innovation and transformation agendas by translating emerging technologies into scalable business opportunities through targeted research and proof-of-value initiatives.
- Lead the design and deployment of scalable AI & IoT platforms, balancing performance, security, interoperability, and long-term sustainability.
- Establish governance frameworks and operating models to ensure compliance, standardization, risk management, and technical excellence across the organization.
- Develop organizational capabilities by mentoring leaders and fostering cross-functional collaboration to accelerate adoption and deliver measurable business impact.

*(T) AI, **Physical AI**, IOT, Industry 5.0, Robotics, Cloud and Hybrid Architecture (AWS and Azure), NVIDIA HPC, Python, AI and Agentic AI, MLOps, Digital Twin, **Amazon Web Services (AWS), Azure,** Pytorch and Tensorflow, Large Language Models (LLM),Docker, Kubernetes, Industrial Internet of Things (IIoT),Scikit-Learn, LLamaIndex, LangChain, MLFlow, SGLang, vLLM, QDrant, SnowFlake,, MCP, RAG, Graph RAG, Digital Twin, **Llama**, BedRock, SageMaker, **DataBricks**, Data Governance, Data Mesh, Data Strategy,*

Digital Services Director (Strategic Technology Program) - Nov 2024 – Feb 2026

Baker Hughes, Florence, Italy

- Delivering digital services in industry, including infrastructure (IoT), applications (Cloud), analytics (AI/ML), and ensuring 24/7 customer support.
- Supporting AI and IoT strategy and develop new digital solutions, while also supporting sales and commercial functions to gain and retain customers.
- Budget responsibility approx. 20M$ Budget. Proven ability to model and communicate the financial and strategic implications of technical decisions to a non-technical audience
- Promoting technical excellence including Industry 4.0, IOT, AI and ML applications.
- Leading a global group (90+ peoples globally EU, USA, APAC), Leading Product Leaders team
- Promoted strategic collaboration with Anybotics and Oracle

(T) AI, IOT, SCADA, HMI, Industry 4.0, Cloud and Hybrid Architecture (AWS and Azure), NVIDIA HPC, Python, Java, Common industrial and automation protocols (OPC-UA, Profinet, Profibus, Modbus, SCADA), Microsoft Azure, Cloud Application Development, Strategic Planning, Soft Skills, Cybersecurity, Generative AI and Agentic AI (LLamaIndex, MPC, A2A, LangChain), AI Management, Cloud computing, ***Amazon Web Services (AWS), EC2, Fargate, BedRock, ECS, ELS, S3, RDS, Azure IoT****, Python, Pytorch and Tensorflow, Large Language Models (LLM), Docker, Business development, Industrial Internet of Things (IIoT),Scikit-Learn, Data Governance, MES, SIEM, PowerBI*

AI Specialists Director (Strategic Technology Program) - Dec 2018 - Nov 2024

Baker Hughes, Florence, Italy

- Developing Machine Learning, Deep Learning, Active Learning and Generative AI (Mistral, ChatGPT, LLAMA, BedRock) products in the field of Energy Transition, Industry 4.0 and Process Automation. Building Business AI Strategy and Competencies,
- Building models: ML, LLM Fine Tuning, Visual Language Model
- Build Cloud Infrastructure: Cloud Foundry, AWS, Docker, Fargate, EC2, redis, DynamoDB, C3.ai - Applications Development:, Spark
- Leading a group of 20 resources globally (EU, APAC)
- Promoted strategic collaboration with NVIDIA, University of Pennsylvania/Siena/PoliMI and Data Bricks

(T) Microservices, Cassandra, Neo4J, Java, Spring, NodeJs, Angular, Cassandra, Kafka, RabbitMQ, Neo4J, gRPC, C/C++, JavaScript – Digital - Transformation and Innovation Leader: Data Lake, S3, HDFS, ***DataBricks****, Github Actions, Agentic AI (LlamaIndex, MCP, LangChain), LLM Fine Tuning (LORA and QLORA), VLM (mondream2),* TensorFlow, Keras, Scikit-Learn, Cuda, Nvidia GPU, GCP TPU, **AWS SageMaker**, Python, DJANGO,Pytorch *Transformation and Innovation Leader: Data Lake, S3, HDFS, Github Actions*

IoT and AI Product Technical Manager - May 2017 - Dec 2018

General Electric, Florence, Italy

- Developing digital products in the field of IIoT - Developing IoT analytics: Predictive Maintenance, Anomaly Detection,
- Production forecasting and optimization - Product Validation of strategies - Demo and Prototype development for customer

(T) Spark, Microservices, Cassandra, Java, Spring, NodeJS, Polymer, Angular, Angular, Hive, Hadoop, Python, Cassandra, Kafka, RabbitMQ, HBase, Cloud computing, Cloud foundry, ***AWS****,* ***Docker, Kubernetes****, Asset Performance Management, IOT, Predix, Industry 4.0, Common industrial and automation protocols (OPC-UA, Profinet, Profibus, Modbus), SIEM*

AI and IoT Technical Leader - Dec 2014 - May 2017

General Electric, Florence, Italy

- Developing Data Processing System
- Developing analytics application, AI and predictive maintenance - Turbomachinery data acquisition - Cybersecurity management - Industrial Internet of Things (IIoT)

(T) Hadoop, Cassandra, Spark, RabbitMQ, Redis, Java, JEE, Oracle, Proficy Historian, Neo4J, GE Predix, Angular, AngularJS, Spring Boot, OPC-UA, Profinet, Profibus, Modbus, Cloud, Python, Anaconda, Polymer, Oauth, NOSQL, AWS, CI/CD

Team Leader- Oct 2012 - Dec 2014

Engineering Group, Florence, Italy

- Developing GPS/DSRC Based Toll System - Big Data Processing and IoT - SOA and ERP Integration, Remote Diagnostic System, Software Quality, Software Management and Delivery
- Scrum, Squad and Team Leader

(T) Java, JEE, javascript, Angular jQuery, C#, C++, .NET, ASP.NET, SQL Server, Biztalk, IIS, Oracle, DB2, PostgreSQL, MySQL, PHP, DROOLS, JSF, RichFaces, Spring, Spring MVC, JBOSS, Glassfish, Tomcat, Portlet, WebSphere, CI/CD

Director of Information Technology - Jan 2007 - Oct 2012

Etruria Innovazione, Siena, Italy

- IT Group Leader (12 resources).
- A//NLP Tech Leader - Architecting and defining technology - NET and Java architect promoting certification and quality system - sector: Public Administation, Public Health, Life Science, BioInformatic, Machine Learning, Business Intelligence
- Promoted collaboration with Universities and Companies

(T) Java, JEE, javascript, jQuery, C#, C++, .NET, ASP.NET, SQL Server, Biztalk, IIS, Oracle, DB2, PostgreSQL, MySQL, PHP, DROOLS, JSF, RichFaces, Spring, Spring MVC, JBOSS, Glassfish, Tomcat, Portlet, WebSphere, VOIP

Software Architect - Jan 2006 - Jan 2007

ION Trading, London, UK and Pisa, Italy

- MarketView Developer and Architect - Developing multithread and real time transaction processing - Developing Platform Console - Developing Platform Security System , Strategizing new products

(T) c/c++, Java, C#, WPF,.NET, QT, Real Time System, Automatic Trading, MarketView

Chief Technology Officer (CTO) - Jan 2004 - Jan 2006

Etruria Innovazione, Siena, Italy

- IT Group Leader (12 resources).
- Product Manager Leader defining products - NLP/AI
- Technical Leader - .NET and JEE architect

(T) Java, JEE, javascript, jQuery, C#, C++, .NET, ASP.NET, SQL Server, Biztalk, IIS, Oracle, DB2, PostgreSQL, MySQL, PHP, SVM, JSF, RichFaces, Spring, Spring MVC, JBOSS, Glassfish, Tomcat, Portlet, WebSphere

President & Cofounder - Jan 2001 - Dec 2004

Metaresearch, Siena, Italy

- Owner and cofounder of Metaresearch srl (BI and NLP start-up).- President - General Manager and CTO
- Product Development and Architect:
- Developing WorkFlow system, CMS and NLP system, BI and ML system, Developing Big Text Data Processing system - Developing Multimedia System

(T) Java, J2EE, Oracle, DB2, ASP 3.0, C/C++, ASP.NET, C#

AI and Computer Vision Developer - Oct 1999 - Jan 2004

University of Siena, Siena, Italy

- Multimedia & DTT Developer - Developing WebTv system - Developing Real Time Big Image Processing system - Developing Biomedical system

(T) C/C++, C#, .NET, Java, MySQL, DB2

EDUCATION

Doctor of Philosophy (PhD) • Neuroscience and Artificial Intelligence • Jan 2008 - Jan 2012

Università di Siena

Master's Degree • Telecommunications Engineering • Jan 1992 - Jan 1999

Università di Siena

CERTIFICATIONS

SAFe 6 Practitioner, SAFe by Scaled Agile, Inc**. Intelligent Video Analytics**, NVIDIA AI. **Green Belt Certified** - Lean Six Sigma International Association of Professionals. **Java OCA Certification** IZ0-803, Oracle. **C3 Fundamentals** C3 AI. **Natural Language Processing Specialization**, Coursera. **Custom Models, Layers, and Loss Functions** with TensorFlow, Coursera. **DataBricks** Ai Agent

ACHIEVEMENTS AND RESEARCH

50+ PAPERS, 3 BOOKS, H-INDEX:10, 330+ CITATIONS, 8 PATENTS, 7 AWARDS, 7 CERTIFICATIONS
Publications: https://scholar.google.it/citations?user=B40SHWAAAAAJ&hl=it
Patents: https://patents.google.com/?inventor=giacomo+veneri&oq=giacomo+veneri

Last Five Publications

1. Knowledge Aware Artificial Intelligence Models for Text Generation in Energy Industry,Fifth International Conference on Computing and Network Communications. CoCoNet 2023.,1221,Lecture Notes in Electrical Engineering,,2025,Springer
2. Agentic AI on Customer Dashboards in Remote Monitoring and Diagnostic Services,Abu Dhabi International Petroleum Exhibition and Conference ,2025,SPE
3. Leveraging the Regularizing Effect of Mixing Industrial and Open Source Data to Prevent Overfitting of LLM Fine Tuning,"International Joint Conference on Artificial Intelligence 2024 Workshop on AI Governance: Alignment, Morality, and Law", 2024,
4. Harnessing Generative AI for Interactive System Failure Diagnostics: A User-Centric Approach to Streamlined Problem Solving and Maintenance,Abu Dhabi International Petroleum Exhibition and Conference,D011S020R006,2024,SPE
5. Enhancing Enterprise-Wide Information Retrieval through RAG Systems Techniques, Evaluation, and Scalable Deployment",Abu Dhabi International Petroleum Exhibition and Conference,D011S020R002,2024,SPE

Books

1. Hands-On Industrial Internet of Things: Build robust industrial IoT infrastructure by using the cloud and artificial intelligence, Packt 2024
2. Hands-On Industrial Internet of Things, Packt 2018
3. Maven Build Customization, Packt 2014

Patents

1. WO-2025153725-A1 - Providing explanations of anomalies in industrial machines or plants through language sentences
2. WO-2025157712-A1 - Determining anomalies on surfaces of or associated to industrial machines or components
3. WO-2024141175-A1 - Root cause analysis of anomalies in turbomachines
4. WO-2023237228-A1 - Method and system for processing analysis tasks regarding computational analyses of industrial systems data
5. WO-2025051392-A1 - Computer-based device implementing a physical model and a data-driven model of a machine
6. IT-202200012071-A1 - Method and system for carrying out evaluations on a plurality of industrial systems
7. IT-202200015669-A1 - Method for generating an output signal from a virtual sensor of an industrial machine with reliability/confidence verification
8. WO-2011001457-A3 - Device for the analysis of the central nervous system by application of a combination of stimuli and the study of corresponding reactions